

Bit Manipulation

$$\begin{array}{r} (11)_{10} \rightarrow 1011 \\ (9)_{10} \leftarrow + 1001 \\ \hline 2010 \\ \hline 2 \times 10 = 20 \end{array}$$

Bit $\rightarrow$ Binary Digit

Bitwise operation

① Binary AND or Bitwise AND

② Binary OR

③ Binary XOR

④ One's complement $\sim$ (not operator)

⑤ Binary left shift $\ll$

⑥ Binary Right shift $\gg$

X	Y	$x \& y$	$x y$	$x \wedge y$
0	0	0	0	1
0	1	0	1	0
1	0	0	1	0
1	1	1	1	1

XOR

$$x \oplus y = x\bar{y} + \bar{x}y$$

X	$\sim X$
0	1
1	0

$$\begin{array}{r} 10 \& 12 = 1010 \\ \& 1100 \\ \hline (1000)_2 = 8 \end{array}$$

$$\begin{array}{r} 10 | 12 = 14 \\ \vee 1010 \\ \vee 1100 \\ \hline (1110)_2 = 14 \end{array}$$

$$\begin{array}{c} \sim (10110110)_2 = (01001001)_2 \xrightarrow[+1]{2^{\text{nd comp}}} (01001010)_2 \\ \xrightarrow{\text{1's comp}} \end{array}$$

$$(10111000)_2 \xrightarrow{2^{\text{nd comp}}} (01001000)_2$$

n bit

the range of +ve integers = $[0, 2^{n-1}-1]$

↓ negative numbers

n=3 $= [-3, 3]$

0	0	0	→ +0
0	0	1	→ +1
0	1	0	→ +2
0	1	1	→ +3
1	0	0	→ -3
1	0	1	→ -2
1	1	0	→ -1
1	1	1	→ -0

$$\begin{aligned} +0 &= 000 \\ -0 &= 111 \end{aligned} \quad \left. \vphantom{\begin{aligned} +0 &= 000 \\ -0 &= 111 \end{aligned}} \right\} 0$$

$$\text{Range [negative + positive]} = [-2^{n-1}, 2^{n-1}-1]$$

↓
[1's complement]

Positive negative
2nd complement

3 bit integer

0	0	0	→ 0
0	0	1	→ 1
0	1	0	→ 2
0	1	1	→ 3
1	0	0	→ -4
1	0	1	→ -3
1	1	0	→ -2
1	1	1	→ -1

3 bit

$$(-4)_{10}$$

$$(4)_{10} = (100)_2$$

↓
2nd comp X

MSB

MSB = 0 → +ve
1 → -ve.

bit size = 3

$$(-3)_{10} = (?)_2$$

$$(3)_{10} = (011)_2 \xrightarrow{2^{nd}} (101)_2$$

↓ 1's comp

$$\sim(3)_{10} = 100$$

$(101)_2$ its negative representation of 3 i.e -3

$$\text{Range} = [-2^{n-1}, 2^{n-1}-1]$$

for 4 bit
Range → $[-8, 7]$

Right shift

$$(25)_{10} \gg 2$$

$$16 + 8 + 1$$

$$(25)_{10} = (0011001)_2 \gg 2$$



↓ ↓
4 2

$$= 6$$

$$(25)_{10} \gg 2 = 6$$

$$25 / 4 = \lfloor 6.25 \rfloor$$

$$\text{num} \gg \text{shift} = \lfloor \text{num} / 2^{\text{shift}} \rfloor$$

Q any Number will be given, finds

- (i) Get i^{th} bit $\rightarrow$ 0
- (ii) Set i^{th} bit $\rightarrow$ new number [i^{th} bit = 1]
- (iii) clear i^{th} bit [i^{th} bit = 0]

$\hookrightarrow$ new number

00010101
3rd bit
3rd bit = 1
→ And the num. (new num)

3rd bit

00010101

00000100

00000100 $\gg 2$

4th bit

00010101

00001000

(00000000)₂ = (0)₁₀

bit = 0

bit mask

i^{th} bit = ?

Magical Num = $1 \ll (i-1)$

Given Number & Magical Number = Result $\gg (i-1)$

0
bit 0
1
bit 1

day-8 integer
 num = 10100100
 2
 bin 10
 (or) 10101100
 set(3)
 10100100
 00001000 ← bit mask (magical num).
 1 << i

$$(00000001)_2 \ll 3$$

$$= (00001000)_2$$

Set bit = num | (1 << i)

Clean ith bit
 10011100
 0 2nd bit ← 3rd position
 and (2) 11111011 ← bit mask.
 bit mask ~ (1 << i)

$$\sim [(00000001)_2 \ll 2]$$

$$\sim (00000100) = 11111011$$

$$\begin{array}{r} 10010010 \\ 11101111 \\ \hline 10000010 \end{array}$$

Clean last i bit

num = 10011011
 ans → 10010000

i = 4

ans = num & [~0 << i]

num = 10011011
 and (2) 11110000 ← bit mask.

bit mask = (~0) << i

$$(\sim 0) = -1$$

$$= 11111111$$

$$-1 \ll 4 = 11110000$$

$$\text{Math.pow}(a, n) = a^n$$

Time complexity $O(n)$

$$O(\log n)$$

$$a^n = \underbrace{a \cdot a \cdot a \dots a}_{n \text{ times}}$$

Fast Exponentiation

$$a = 3^9 = 3 \cdot 3^8 = 3 \cdot (3^2)^4 = 3 \cdot (9)^4 = 3 \cdot (9^2)^2 = 3 \cdot (81)^2$$

$$= 3 \cdot (6481)^1$$

$$= 3 \cdot (6481) \cdot (6481)^0 = 3 \cdot 6481 \cdot 1$$

ans = 1
while (n > 0) {
if (n % 2 == 1) {

$$n = n - 1$$

{
else { // even

$$n = n / 2$$

$$a = a * a$$

}

$$ans = ans * a ;$$

}

$$a = 3$$

$$n = 9 = 1001$$

$$ans = 3$$

$$a = 3 \times 3 = 9$$

$$n = 0100$$

$$n = 0100 \Rightarrow a = 9 \times 9 = 81, n = 0010$$

$$n = 0010 \Rightarrow a = 81 \times 81 = \dots, n = 0001$$

$$n = 0001 \Rightarrow ans = \underbrace{81 \times 81 \times 3}_a \uparrow ans$$

$$3^2 = 9$$

$$3^4 = (3^2)^2 = 9 \times 9 = 81$$

$$3^8 = (3^4)^2 = 81^2 = 6481$$

$$3^9 = 3 \cdot 3^8 = 3 \cdot 6481$$

using bit manipulation

$$\text{int ans} = 1 ;$$

while (n > 0) {

if ((n & 1) != 0) {

$$ans = ans * a ;$$

}

$$a = a * a ;$$

$$n \gg 1 ;$$

}

return ans ;

~~9.10~~ $2^{10} = \underbrace{2 \cdot 2 \cdot 2 \cdot \dots \cdot 2}_{10 \text{ times}} = O(10) = O(m)$

$a = a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a$
 $2^4 \cdot 2^1 \cdot 2^2 \cdot 2^1 \cdot 2^0$
 $1001 = (3^8 \times 1) \cdot (3^4 \times 0) \cdot (3^2 \times 0) \cdot (3^1 \times 1)$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $(3^4)^2 \quad (3^2)^2 \quad 3 \times 3 \quad 3$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $81 \times 81 \quad 9 \times 9 \quad 9 \quad 3$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $81 \quad 81 \quad 9 \quad 3$

$n \xrightarrow{\text{bit}} \lceil \log n \rceil \text{ length of bit stream}$

$T.C = O(\log n)$

Ex ***
Modular Exponentiation

$a^n \cdot x = 0, 1, 2, \dots, m-1$

$3^5 \text{ mod } 7 = 3 \cdot 3^4 \text{ mod } 7$
 $= 3 \cdot (3^2)^2 \text{ mod } 7$
 $= 3 \cdot 9^2 \text{ mod } 7 = 3 \cdot 4 \text{ mod } 7 = 12 \text{ mod } 7 = 5$

$\therefore \{ 9 \text{ mod } 7 = 2 \}$
 $\therefore 9^2 \text{ mod } 7 = 2^2 \text{ mod } 7 = 4 \text{ mod } 7 = 4$

$\uparrow$
 inter. step.
 $3 \cdot 9 \text{ mod } 7 = 12 \text{ mod } 7 = 5$
 if, $a \text{ mod } m = x \in \{0, 1, \dots, m-1\}$
 then $a^n \text{ mod } m = x^n \text{ mod } m$
 if $a \text{ mod } m = x$
 $b \text{ mod } m = y$
 then $a \cdot b \text{ mod } m = x \cdot y \text{ mod } m$